

SUPERSTORE SALES ANALYSIS

End-to-End SQL Data Cleaning & Business Intelligence Project

Prepared by

M SureshKumar

SQL / Data Analyst Candidate

Tools Used	PostgreSQL, pgAdmin, SQL (CTEs, Window Functions, Subqueries, CASE WHEN)
Dataset Size	9,994 Retail Transactions
Date	June 2026

1. EXECUTIVE SUMMARY

This project demonstrates an end-to-end data analysis workflow using SQL. Starting from raw CSV data, a database schema was designed, rigorous data validation and cleaning was performed, and a multi-level business analysis was conducted to uncover profitability drivers.

Key Achievements

- Data Governance: Cleaned 9,994 records by identifying and removing 17 duplicate entries.
- Root Cause Analysis: Discovered that the Central Region underperformed due to excessive discounting (24% average) on high-cost items.
- Critical Insight: Identified "Tables" as a loss-making product (-8.56% margin) despite moderate discounts, proving that product cost is a more critical factor than discount alone.
- Business Impact: Delivered actionable recommendations to improve profit margins by approximately 15%.

2. PROJECT OVERVIEW

The goal was not just to run queries but to think like a Business Analyst:

- Clean and validate raw data.
- Design a proper database schema.
- Perform structured drill-down analysis.
- Identify root causes of profit loss.
- Deliver strategic business recommendations.

This project simulates a real-world retail business scenario where data quality directly impacts financial decisions.

3. BUSINESS PROBLEM

The company wanted to understand:

1. What is the overall health of the business?
2. Which regions generate the highest and lowest profit?
3. Why is one region underperforming?
4. Are discounts helping or hurting the business?
5. Which products should be promoted or controlled?

4. DATASET DESCRIPTION

The dataset contains 9,994 retail sales transactions across the following dimensions:

- Geographic: Country, State, City, Region
- Customer Segment: Consumer, Corporate, Home Office
- Product: Category, Sub-Category
- Financial: Sales, Quantity, Discount, Profit

5. DATABASE DESIGN

A structured table named 'store' was created using proper data types to ensure data accuracy.

Why Proper Data Types Matter

- VARCHAR: Used for text and postal codes (to preserve leading zeros).
- DECIMAL(10,2): Used for financial values (sales, profit, discount) to avoid rounding errors.
- INTEGER: Used for quantity (whole numbers).
- SERIAL PRIMARY KEY: Used for unique row identification.

SQL Schema Design

```
CREATE TABLE store (  
    row_id          SERIAL PRIMARY KEY,  
    ship_mode       VARCHAR(20),  
    segment         VARCHAR(20),  
    country         VARCHAR(50),  
    city            VARCHAR(50),  
    state           VARCHAR(50),  
    postal_code     VARCHAR(10),  
    region          VARCHAR(20),  
    category        VARCHAR(50),  
    sub_category    VARCHAR(50),  
    sales           DECIMAL(10,2),  
    quantity        INTEGER,  
    discount        DECIMAL(4,2),  
    profit          DECIMAL(10,2)  
);
```

This ensures professional-level database integrity and performance.

6. DATA CLEANING AND VALIDATION

Before analysis, a 7-step validation process was performed to ensure data quality.

Step 1: Missing Value Check

- Checked NULL values in all numeric columns.
- Checked empty strings using TRIM() function.
- Result: No missing values found.

Step 2: Logical Validation

- Sales > 0 — Passed
- Quantity >= 1 — Passed
- Discount between 0 and 1 — Passed

- No invalid profit anomalies — Passed

Step 3: Duplicate Detection

- Grouped all business columns to find exact duplicates.
- Result: Found 17 duplicate records.

Step 4: Safe Cleaning Strategy

Instead of deleting directly from the original table, a safe approach was followed:

6. Created a new clean table called store_clean.
7. Inserted only distinct (unique) rows.
8. Preserved original data as backup.

Validation Logic (SQL Snippet)

```
-- Calculate total extra duplicate rows
WITH dup_counts AS (
  SELECT COUNT(*) AS duplicate_count
  FROM store
  GROUP BY ship_mode, segment, country, city, state,
           postal_code, region, category, sub_category,
           sales, quantity, discount, profit
  HAVING COUNT(*) > 1
)
SELECT SUM(duplicate_count - 1) AS total_duplicate_rows_to_remove
FROM dup_counts;
-- Result: 17 duplicates found

-- Create Clean Table and Insert Distinct Rows
DROP TABLE IF EXISTS store_clean;
CREATE TABLE store_clean (...schema...);

INSERT INTO store_clean (...)
SELECT DISTINCT ... FROM store;
-- Verification: 9,977 unique rows (9,994 - 17)
```

7. BUSINESS ANALYSIS (5-LEVEL DRILL DOWN)

A structured top-down approach was used to move from high-level metrics to root causes.

Level 1: Overall Business Health

- Total Sales: Rs.22.96 Lakhs
- Total Profit: Rs.2.86 Lakhs
- Profit Margin: 12.46%

Observation: Business is profitable, but there is room for margin improvement.

Level 2: Regional Performance

Region	Profit Margin	Avg Discount	Performance
West	14.94%	10.9%	Best
East	13.49%	14.5%	Good
South	11.94%	14.7%	Average
Central	7.92%	24.0%	Critical

KEY INSIGHT: Central region has the lowest margin due to excessive discounting at 24%. This is more than double the discount rate of the best-performing West region.

Level 3: Deep Dive – Central Region

To find the root cause within the Central region, analysis was drilled down by category and sub-category:

- Furniture category had the highest discount at 29.8%.
- Furnishings sub-category had a massive 40.4% discount.

Conclusion: Heavy and uncontrolled discounting is directly impacting profitability in this region.

Level 4: Product Cost vs. Discount Analysis

A critical business insight emerged when comparing two products:

Product	Discount	Margin	Status
Tables	26.1%	-8.56%	LOSS
Binders	37.0%	+14.86%	PROFIT

KEY LEARNING: High discount does NOT always equal loss. High product cost PLUS discount EQUALS loss. Tables are expensive with thin margin space — discounts create losses. Binders are low-cost, so even 37% discount remains profitable.

Level 5: State-Level Root Cause

Drilling down to state level within the Central region revealed the specific culprit:

- Worst Performing State: TEXAS
- Profit Margin: -15.14% (Negative)
- Average Discount: 37.1%

Actionable Finding: Texas is the single biggest driver of Central region's poor performance. The combination of high discounts on expensive products in Texas is creating significant losses.

8. ADVANCED SQL TECHNIQUES DEMONSTRATED

This project showcases advanced SQL capabilities required for senior analyst roles.

Skill 1: Top 3 Products per Region (Window Functions & CTE)

Purpose: Identify regional best-sellers to avoid "one-size-fits-all" marketing.

```
WITH regional_rankings AS (  
    SELECT  
        region,  
        sub_category,  
        SUM(sales) AS total_sales,  
        RANK() OVER (PARTITION BY region  
                     ORDER BY SUM(sales) DESC) AS rank_in_region  
    FROM store_clean  
    GROUP BY region, sub_category  
)  
SELECT *  
FROM regional_rankings  
WHERE rank_in_region <= 3;
```

Skill 2: Above-Average Performers (Nested Subqueries)

Purpose: Find states performing better than the national average.

```
SELECT state, SUM(sales) AS total_sales  
FROM store_clean  
GROUP BY state  
HAVING SUM(sales) > (  
    SELECT AVG(state_avg)  
    FROM (  
        SELECT SUM(sales) AS state_avg  
        FROM store_clean  
        GROUP BY state  
    ) AS avg_calc  
);
```

Skill 3: Profit Classification (CASE WHEN)

Purpose: Categorize products into business-friendly labels for quick decision making.

```
CASE  
    WHEN (SUM(profit)/SUM(sales)*100) > 15  
        THEN 'High Margin (>15%)'  
    WHEN (SUM(profit)/SUM(sales)*100) BETWEEN 5 AND 15  
        THEN 'Medium Margin (5-15%)'  
    ELSE 'Low Margin (<5%)'  
END AS margin_status
```


9. BUSINESS RECOMMENDATIONS

Based on the complete analysis, the following strategic actions are recommended:

9. Reduce Discounts in Central Region: Specifically target Texas. Reduce discounts from 37% to below 15%. This single change can recover significant profit.
10. Product-Level Strategy: Stop aggressive promotion of "Tables" without price adjustments. If base costs cannot be lowered, stop discounting below 10%. Maintain higher discounts only on low-cost items like Paper and Labels.
11. Region-Specific Pricing Policy: Do not apply uniform discounts across all regions. High-cost products should receive low or zero discount; low-cost products can have flexible higher discounts.
12. Inventory Optimization: Stock Chairs and Phones heavily in West/East regions; prioritize Binders in Central region.

10. BUSINESS IMPACT

If the above recommendations are implemented:

- Overall profit margin can improve from 12.46% to over 15%.
- Loss-making regions (Central) can become profitable.
- Smarter discount strategy will protect high-cost product margins.
- Texas alone can shift from -15% loss to breakeven or profit.

11. CONCLUSION

This project successfully transformed raw retail data into actionable business intelligence. The core issue identified was:

"Excessive discounting on high-cost products is the primary cause of profitability damage."

By combining technical SQL skills (data validation, window functions, CTEs, subqueries) with business acumen, strategic recommendations were delivered that could significantly improve company performance.

About the Author

M SureshKumar

Aspiring Data Analyst | SQL Specialist

Focused on transforming raw data into meaningful business decisions.